

COMSATS University Islamabad (CUI)

Software Design Description

(SDD DOCUMENT)

for

PodVerse

By

Rabeea Chughtai SP22-BCS-079/ISB

Sadia Imran SP22-BCS-087/ISB

Supervisor

Bachelor of Science in Computer Science (2022-2026)

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1. Introduction

PodVerse is an advanced AI-powered podcast system crafted to simplify and enhance the podcasting experience for small business owners, entrepreneurs, and content creators. Designed to be user-friendly yet feature-rich, PodVerse enables users to create, manage, and interact with professional-quality podcasts effortlessly, making it a perfect tool for sharing stories, promoting products, and connecting with audiences on a deeper level.
The platform offers a wide range of tools and functionalities tailored to both novice and experienced podcasters. It streamlines the entire podcasting process, from ideation to publishing, ensuring that creators can focus on their content without being burdened by technical complexities.

The following are the functionalities of PodVerse that we have covered till now:

- Podcast Creation & Publishing: Allows users to record and edit podcasts with customized AI hosts or guests. Provides export options and saving of recordings for future use.

- AI Host Interaction: Enables users to choose between a general AI Host or an AI Interviewer. Facilitates smooth transitions between podcast segments, tracks time, and provides recap prompts to ensure engaging content flow.

2. Design methodology and software process model

2.1 Software Process Model

For the development of **PodVerse**, we are using the Incremental Software Process Model, which enables the system to be built in manageable phases. Foundational features are implemented first, followed by gradual additions like user authentication, podcast creation, and AI interactions. This approach ensures early delivery of a functional system, facilitates validation and feedback at each stage, and allows seamless incorporation of modifications, reducing risks and improving flexibility.

2.2 Design Methodology

The design methodology used for PodVerse is Procedural Design Methodology. This approach focuses on breaking down the system into smaller, well-defined procedures or modules, each responsible for a specific functionality. It is well-suited for systems with clear and structured workflows like PodVerse, where tasks such as data processing, AI interactions, and content management can be effectively handled in separate, logical procedures.

Advantages of Procedural Design Methodology:

- **Simplicity:** Each task or functionality is clearly defined and implemented as a separate procedure, making the development process straightforward.
- **Ease of Debugging**: With functions divided into smaller, manageable pieces, identifying and fixing bugs becomes easier.
- **Scalability:** As the system evolves, additional features can be integrated into the existing procedures or added as new ones without disrupting the core structure.
- **Efficiency:** Focused procedures ensure optimal utilization of resources and streamlined performance.

3. System overview

PodVerse is an innovative AI-powered podcast platform designed to simplify podcast creation, interaction, and management for small business owners, entrepreneurs, and content creators. It offers a seamless way to create, customize, and publish engaging podcasts while incorporating advanced AI-driven features to enhance user experience.

The platform allows users to create AI-powered hosts and guests, integrate real-time language translation, and access interactive tools for audience engagement. Key functionalities include user authentication, knowledge base setup, voice cloning, podcast creation and publishing, AI-hosted conversations, and analytics to optimize content performance.

With a user-friendly interface and robust backend, PodVerse caters to users of all technical skill levels, offering a streamlined approach to producing professional-quality podcasts. The system also supports customizable features and real-time 3D avatars for immersive podcast interactions. Designed to empower small businesses and creators, PodVerse leverages AI to make podcasting accessible, efficient, and impactful.

3.1 Architectural design

For PodVerse, we use the Layered Architecture to organize the system into three main layers: User Interface (UI), Processing, and Data.

- **UI Layer**: Manages user interaction, including authentication and podcast creation. It communicates with the dashboard and other modules.
- **Processing Layer**: Handles the core logic, such as AI interactions and real-time translation. It processes data using the knowledge base and translator modules.
- **Data Layer**: Responsible for storing and retrieving data from the database, supporting the AI modules and analytics.

This architecture allows each layer to function independently, making the system easier to maintain, scale, and modify as needed.
Here's the Box and Line Diagram illustrating the system's structure:

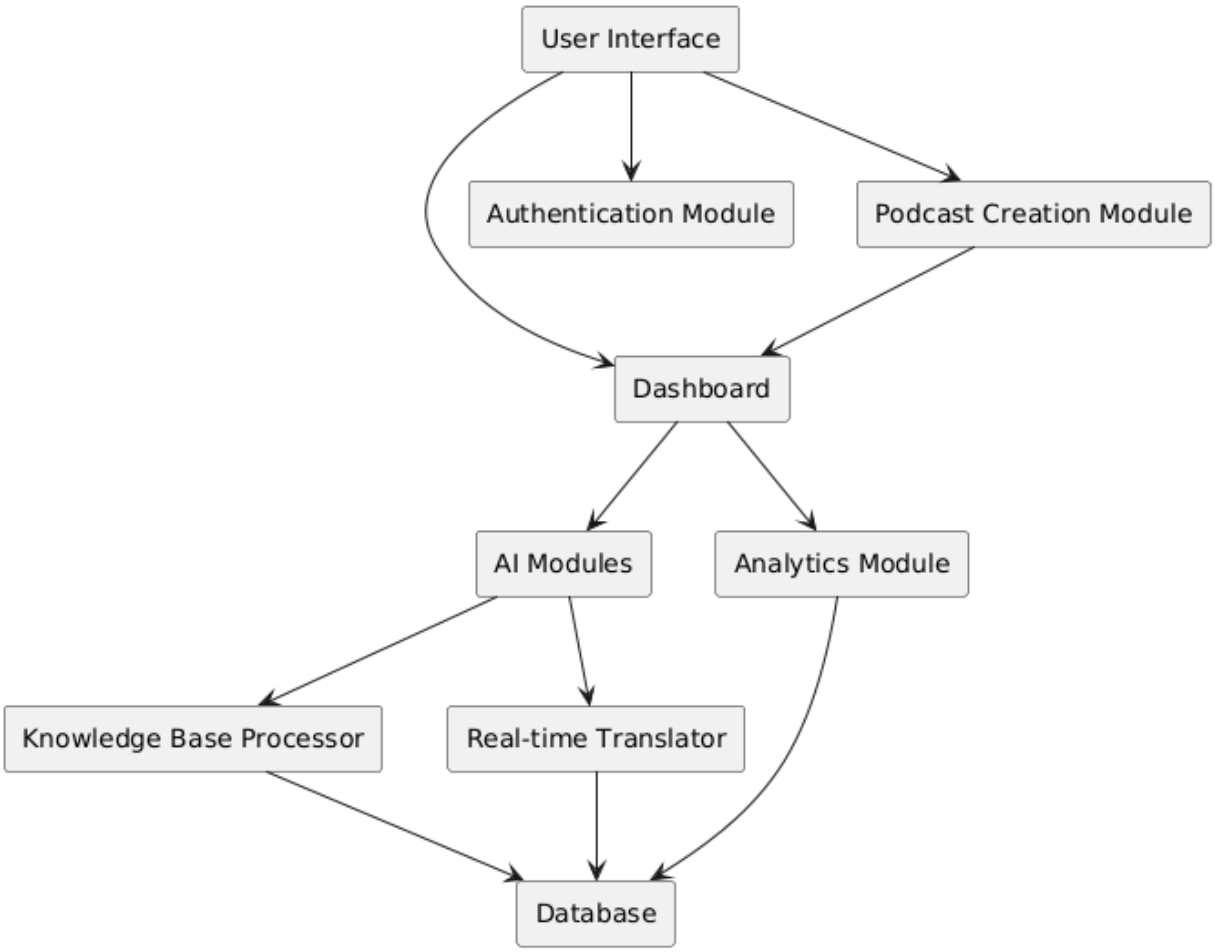


Figure 3.1 Box and Line Diagram for PodVerse

4. Design models

The applicable model of our project includes:

- Activity Diagram
- Data Flow Diagram
- State Transition Diagram

4.1 Activity Diagram

4.1.1 Create Podcast

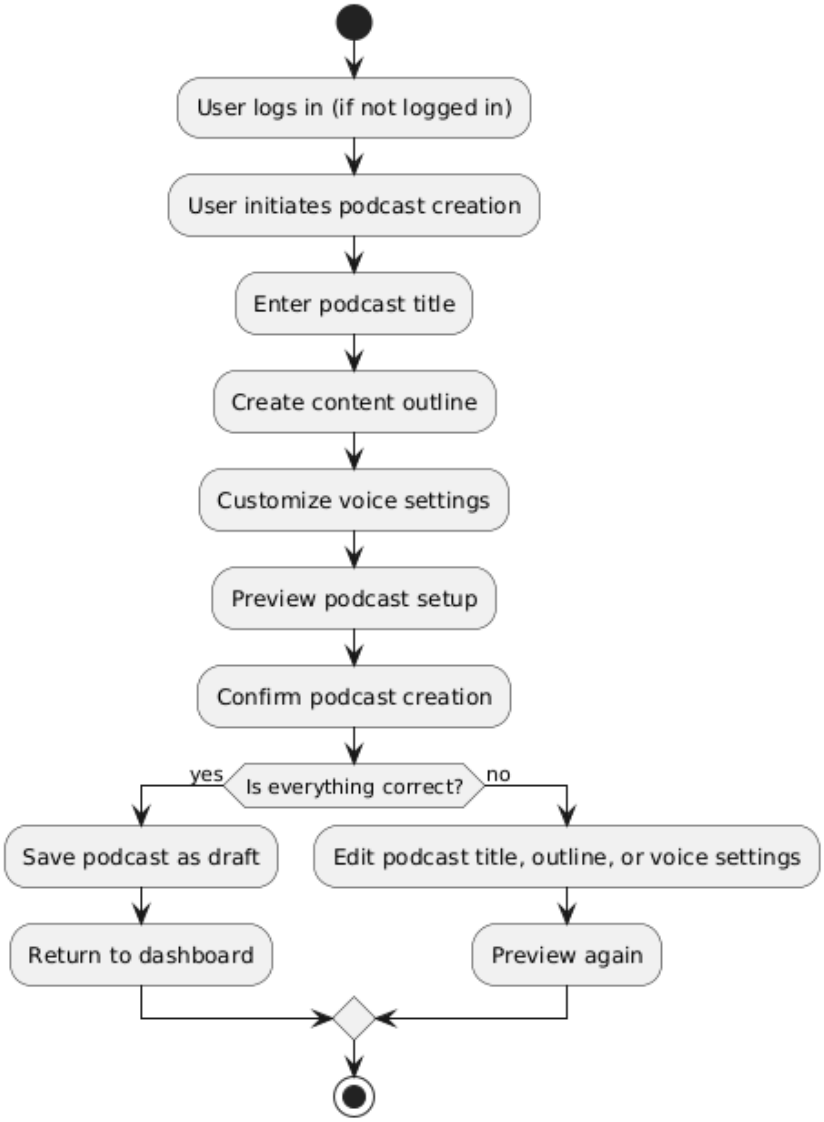


Figure 4.1.1 Activity Diagram of Create Podcast

4.1.2 Edit Podcast

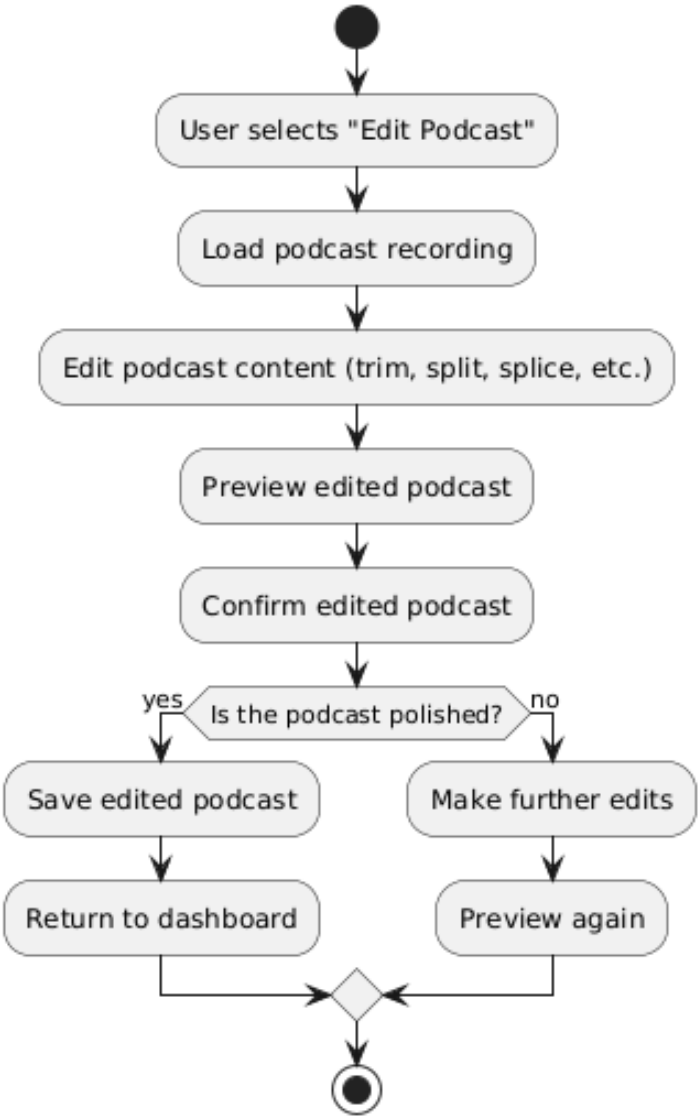


Figure 4.1.2 Activity Diagram of Edit Podcast

4.1.3 Publish Podcast

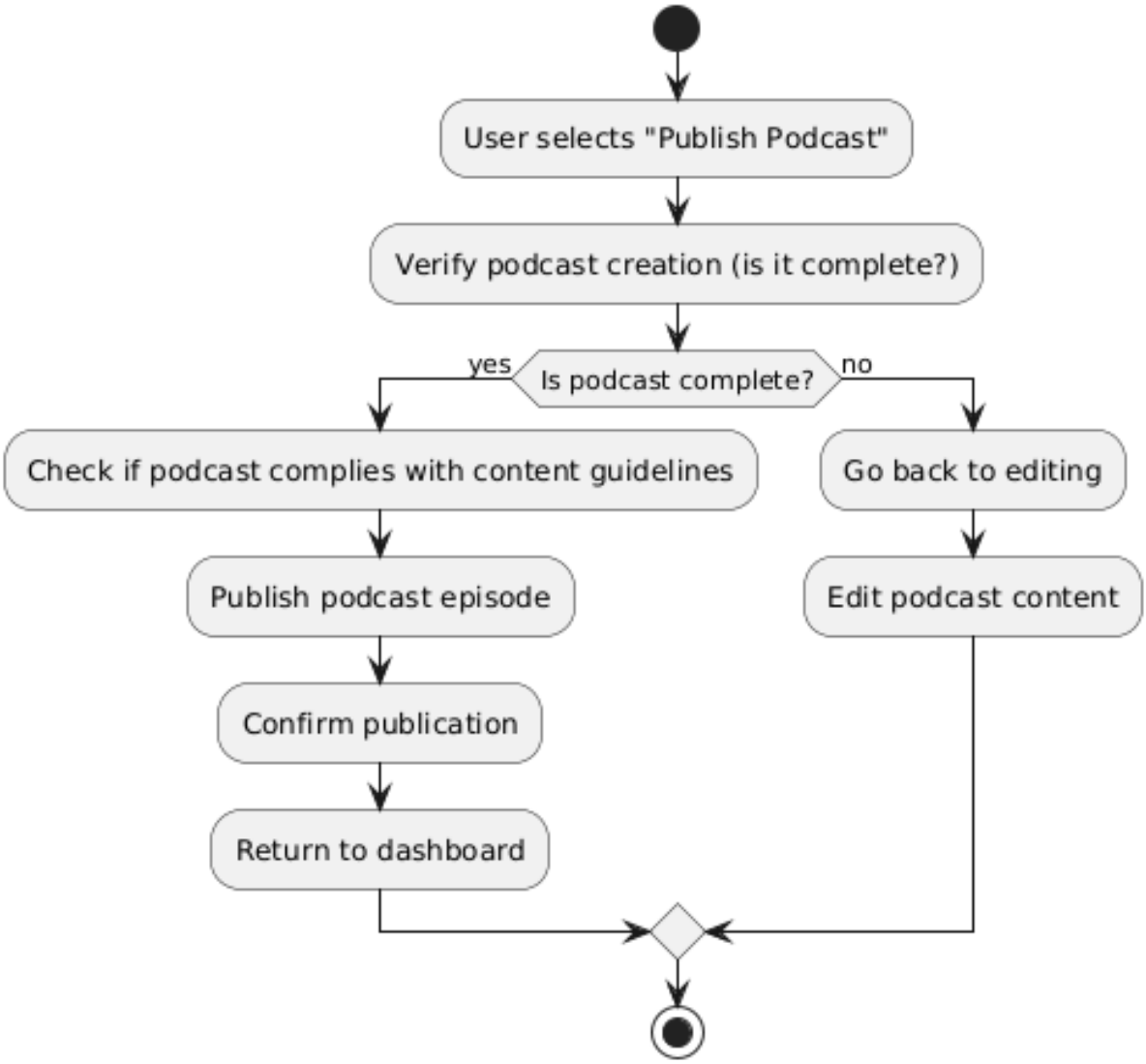


Figure 4.1.3 Activity Diagram of Publish Podcast

4.1.4 AI Host Session Initiation

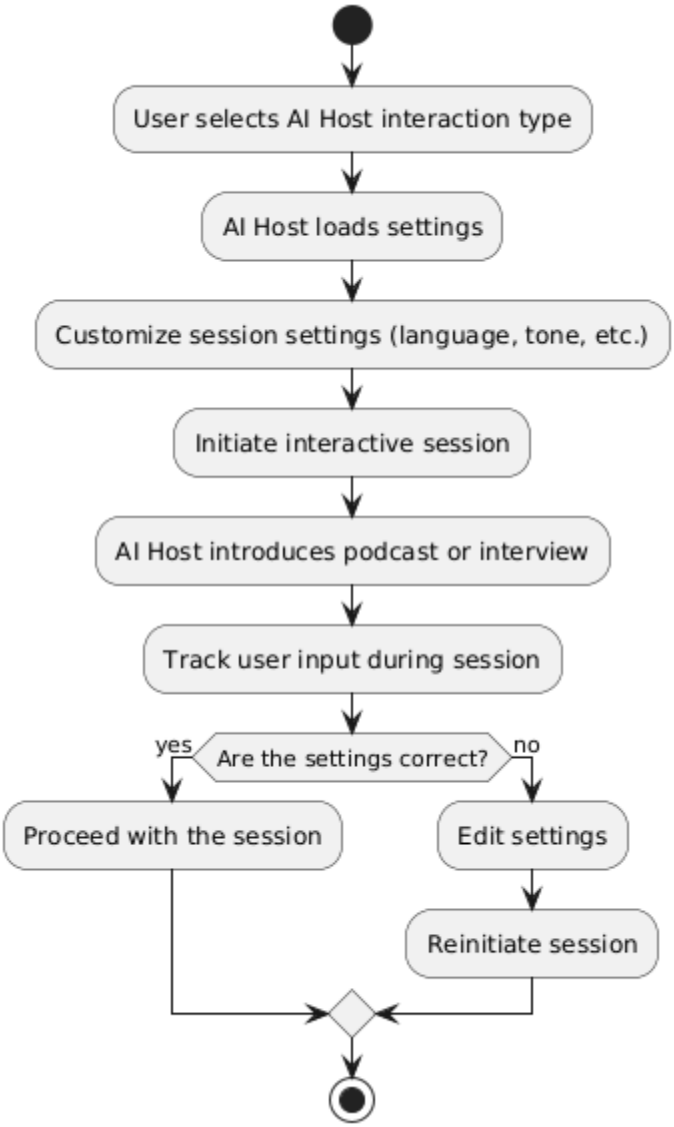


Figure 4.1.4 Activity Diagram of AI Host Session Initiation

4.1.5 Language Translation

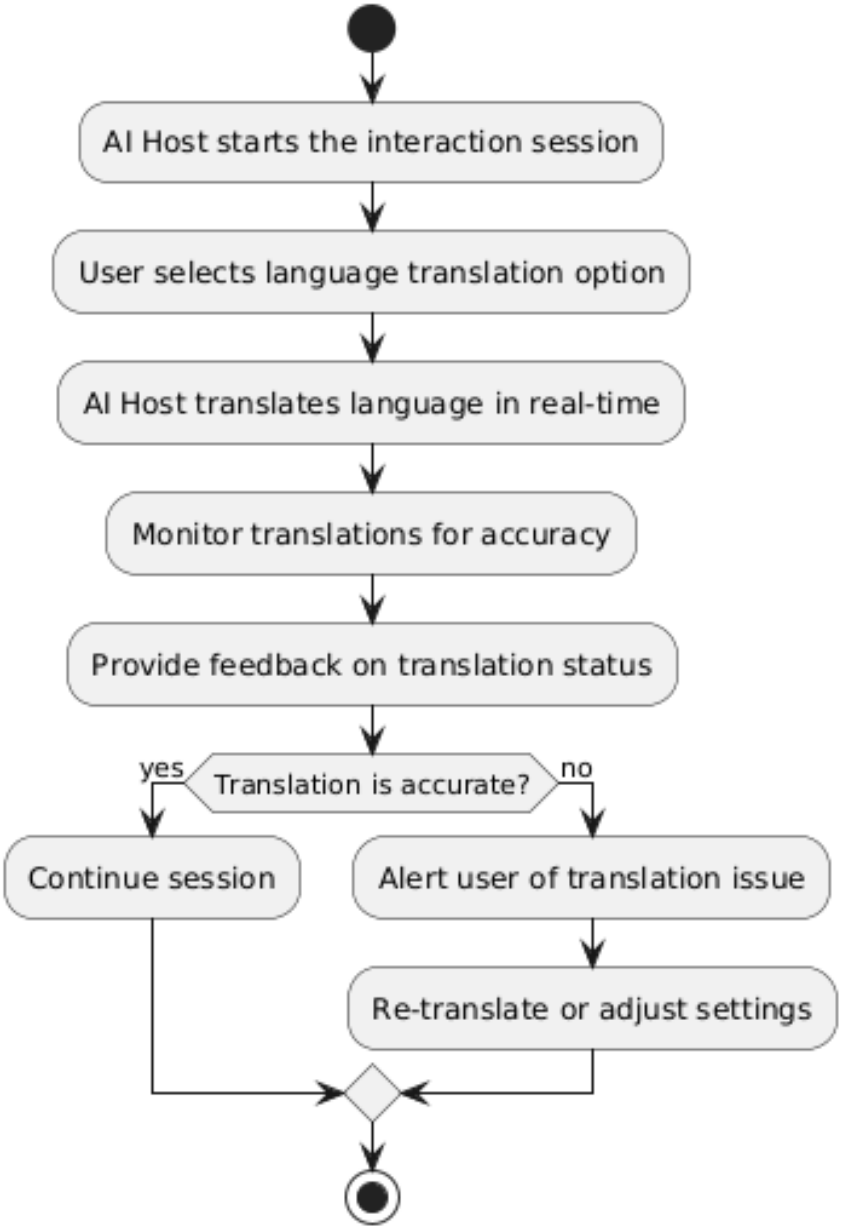


Figure 4.1.5 Activity Diagram of Language Translation

4.1.6 Feedback

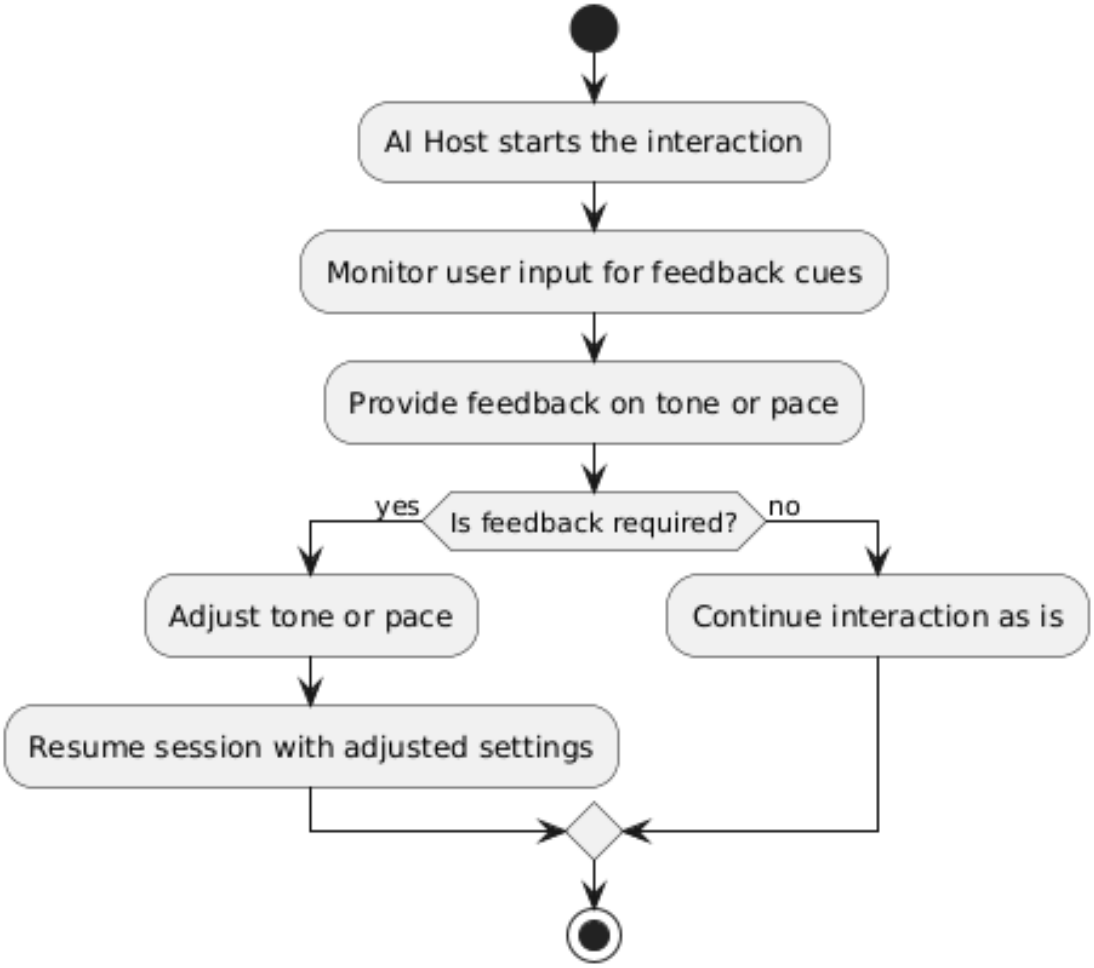


Figure 4.1.6 Activity Diagram of Feedback

4.2 Data Flow Diagram

4.2.1 Context Diagram

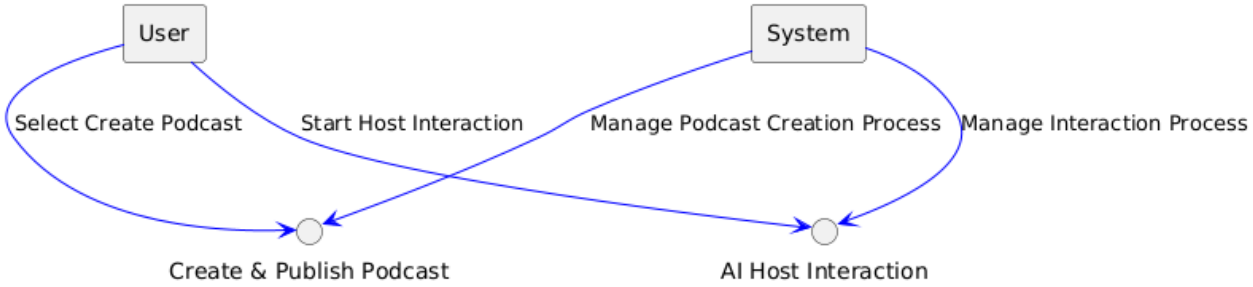


Figure 4.2.1 Context Diagram DFD of PodVerse System

4.2.2 Diagram 0 DFD

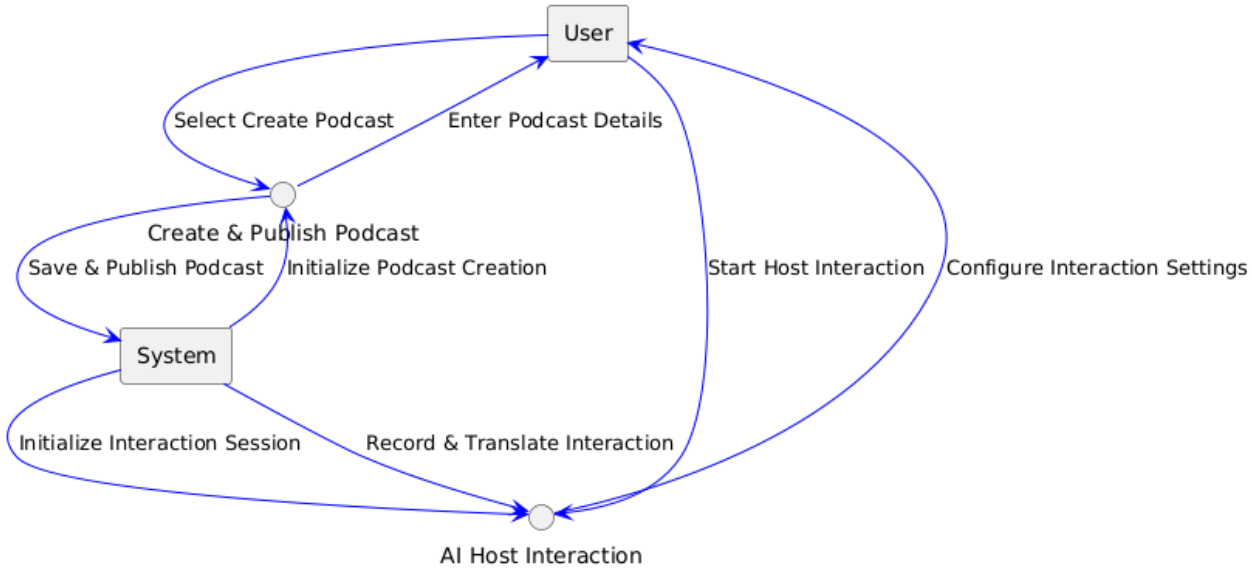


Figure 4.2.2 Diagram 0 DFD of PodVerse System

4.2.3 Diagram 1 DFD



Figure 4.2.3 Diagram 1 DFD for Create and Publish Podcast

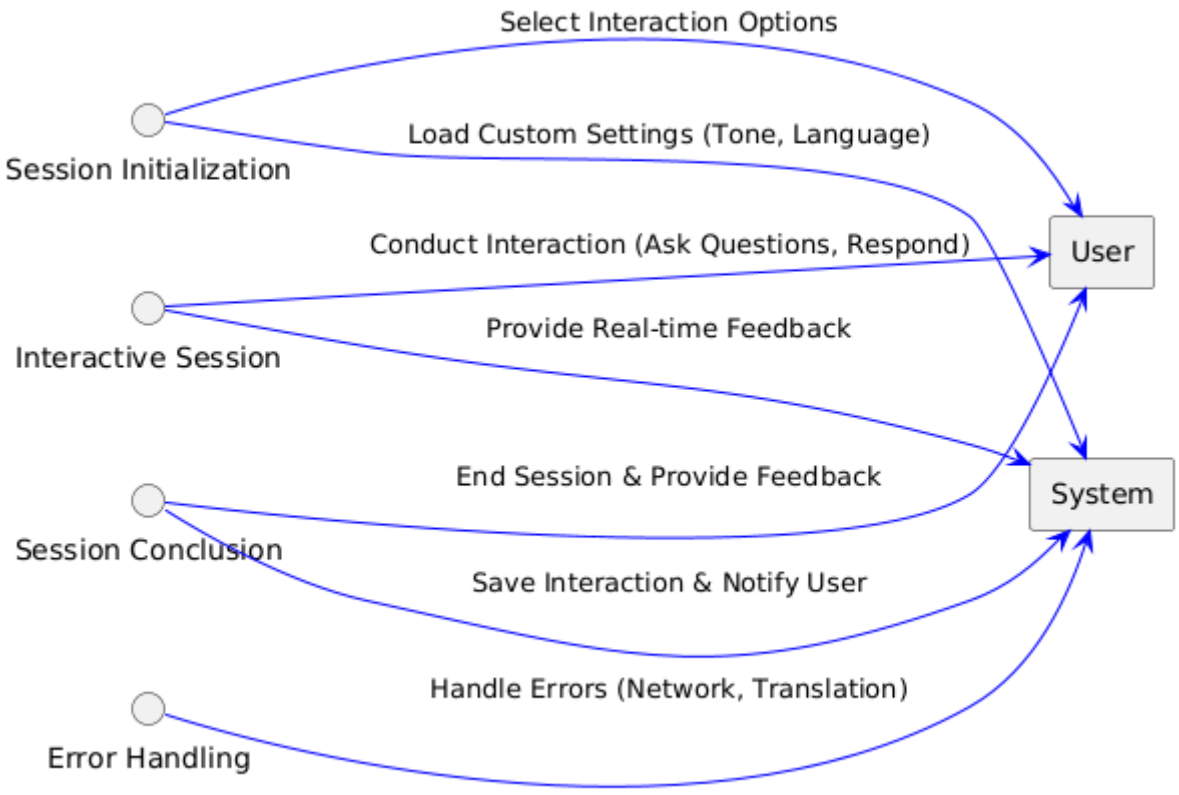


Figure 4.2.4 Diagram 1 DFD for AI Host Interaction

4.3 State Transition Diagram

4.3.1 Create and Publish Podcast

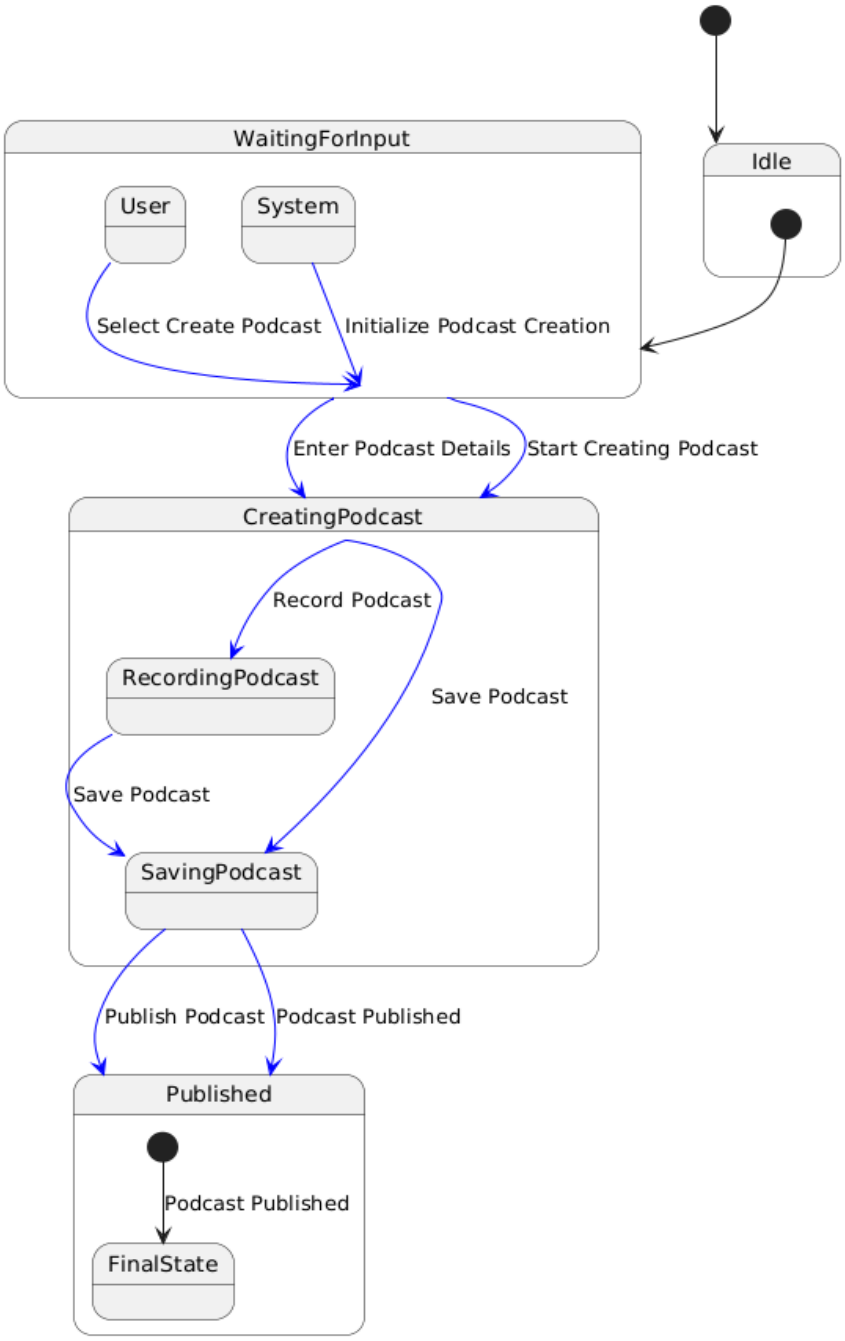


Figure 4.3.2 State Transition Diagram of Create and Publish Podcast

4.3.3 AI Host Interaction

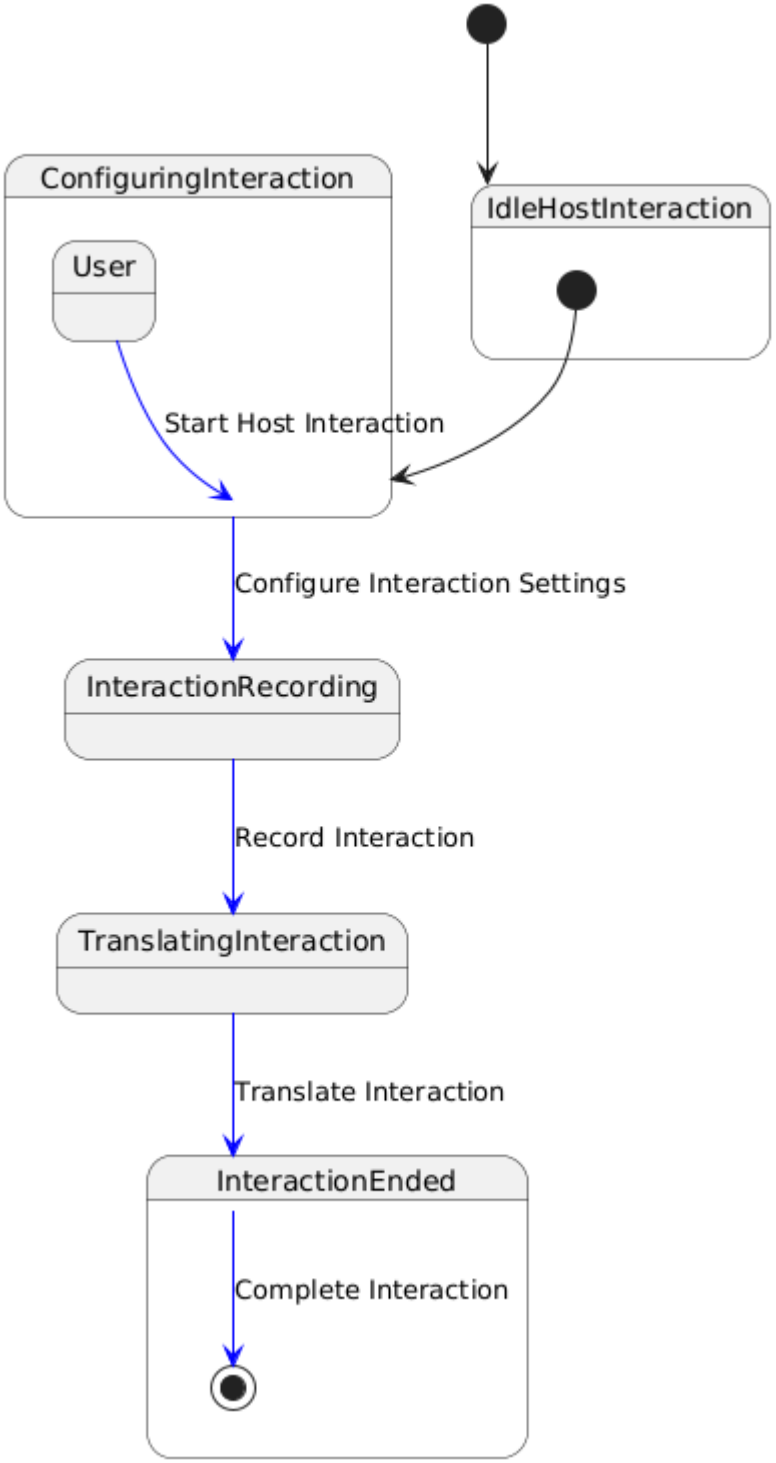


Figure 4.3.3 State Transition Diagram of AI Host Interaction